

An infrared triangle for quantum spin chains

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The infrared triangle of gauge theory and gravity—soft theorem, asymptotic symmetry, memory effect—has no lattice formulation. We give it one, in quantum spin chains. We prove that a soft magnon of the Heisenberg ferromagnet decouples with the universal phase $2k$; that the symmetry behind this acts at spatial infinity as operators on matrix-product bond space, with central extension the SPT index; and that a magnon crossing a domain wall drags it by exactly two sites—a memory effect with its own Adler zero. Bethe-ansatz, exact-diagonalization, and wavepacket numerics verify each claim.

A ferromagnetic spin chain contains two objects that can be pictured without formalism: the *magnon*, a single overturned spin spreading as a wave with energy $\omega(k) = J(1 - \cos k)$, and the *domain wall*, the boundary between a region of up spins and a region of down spins. This Letter concerns what each does when the magnon’s momentum k is taken small, and the symmetry bookkeeping that controls both. The whole result in one sentence: on the lattice, a soft magnon decouples from scattering with the universal phase $2k$; the symmetry responsible for this acts at the two ends of the chain as explicit operators on the bond space of a matrix product state; and a magnon that crosses a domain wall drags it backward by exactly two lattice sites, so the wall keeps a quantized memory of every transmission.

In continuum field theory these three phenomena—soft theorems, asymptotic symmetries, memory effects—form the *infrared triangle*, three faces of one Ward identity, established for photons and gravitons at null infinity [1, 2]. Hamada and Sugishita extended the triangle to Goldstone bosons of a spontaneously broken global symmetry [3]; scalar and fracton versions followed [4–6], together with local derivations that avoid null infinity [7] and a careful account of what infrared finiteness requires [8]. Each ingredient separately also has a condensed-matter ancestor. Dyson showed in 1956 that ferromagnetic magnons decouple at long wavelength [9]; the Adler zeros of nonrelativistic Goldstone effective theories are by now systematic [10–15]; and magnon-driven domain-wall motion is a working subject in magnonics, where Lan and Xiao described the wall’s displacement as a gravity-like two-body collision [16–18]. What has been missing is any statement on the lattice: the infrared-structure literature and the tensor-network literature do not intersect. This Letter supplies the lattice statement, at finite spacing, with no continuum limit and no expansion parameter, and verifies each corner against exact and numerical computation.

Setting.—An infinite chain of spin- s sites; a finite-range Hamiltonian H ; an on-site symmetry group G acting by $u(g)$ at every site; and a translation-invariant injective matrix product ground state with tensor A_{ij}^σ and bond (virtual) dimension χ . One structural fact is used

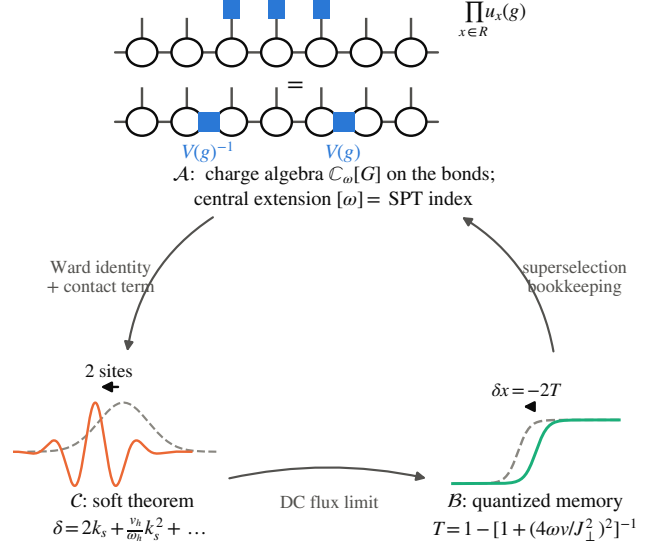


FIG. 1. The infrared triangle at finite lattice spacing. Corner $\mathcal{A}$ (asymptotic symmetry): a symmetry applied to a region of a matrix product state collapses to two bond operators $V(g)^{-1}, V(g)$ at the region’s ends (Theorem WI); sent to $\pm\infty$, these generate the asymptotic charge algebra $C_\omega[G]$, whose central extension $[\omega]$ is the SPT index of the chain. Corner $\mathcal{C}$ (soft theorem): a soft magnon decouples with universal phase $2k_s$, i.e. it exits scattering displaced by two lattice sites (Theorem 1). Corner $\mathcal{B}$ (memory): a magnon transmitted through a domain wall displaces it by exactly two sites; a reflected one by zero (Theorem 2). The edges are the derivations reported in the text: the charge algebra fixes the soft coefficient through a current matrix element; the soft data fixes the memory through the transmission probability alone; the memory is valued in the vacuum-pair data of the broken asymptotic symmetry.

throughout, the fundamental theorem of MPS symmetries [19, 20]: if the state is G -invariant, then acting with $u(g)$ on the physical index of A equals a phase times $V(g)^{-1}AV(g)$, where $V(g)$ is a projective representation on the bond space whose class $[\omega] \in H^2(G, U(1))$ is the symmetry-protected topological (SPT) index of the phase [21, 22]. Excitations are momentum superpositions of local tensor replacements—the tangent-space

ansatz [23–25]—whose S -matrix is computable on the lattice [26, 27]. All theorems below are proved in the Supplemental Material; each carries a machine-checked numerical certificate, and we quote the checked precision in place.

Asymptotic symmetry.—Apply the symmetry to a finite region R only: $U_R(g) = \prod_{x \in R} u(g)$. Using the intertwining relation once per site, the V ’s cancel pairwise throughout the interior of R , and the whole operation collapses to two bond operators, $V(g)^{-1}$ on the bond entering R and $V(g)$ on the bond leaving it. Everything a truncated symmetry does, it does at its two ends. Sending the ends to $\pm\infty$ turns this into an asymptotic-symmetry statement, and three facts hold. (i) On states, a half-infinite symmetry string acts *exactly* as a single bond insertion; as an operator sequence it converges strongly if and only if $V(g)$ is a phase, so whenever the symmetry acts nontrivially on the bond space there is no strongly convergent implementer at all—the lattice counterpart of a large gauge transformation, distinguished from a global symmetry by its failure to converge. (ii) The bond implementers compose as the twisted group algebra $\mathbb{C}_\omega[G]$: the SPT class is precisely the central extension of the asymptotic charge algebra. (iii) If the symmetry is spontaneously broken, the half-infinite string leaves the ground-state sector altogether: it converges (in expectation values) to a *domain-wall state*. Kinks are what broken asymptotic symmetries create, and the kink superselection sectors are labelled by ordered pairs of vacua. On the lattice, an asymptotic symmetry is not a fall-off condition; it is a concrete algebra of $\chi \times \chi$ matrices. This reading of the MPO-symmetry technology [28, 29] is, to our knowledge, new.

The soft theorem.—Now fix the concrete model: the spin- $\frac{1}{2}$ Heisenberg ferromagnet $H = -J \sum_x (\mathbf{S}_x \cdot \mathbf{S}_{x+1} - \frac{1}{4})$. Its fully polarized state is an exact product ground state ($\chi = 1$), magnons are exact one-particle eigenstates with $\omega(k) = J(1 - \cos k)$ and velocity $v(k) = J \sin k$, and $SU(2)$ is broken to $U(1)$, so the magnon is a type-B Goldstone mode [10]. Write $S_{12}(k_s, k_h) = e^{i\delta}$ for the exact two-magnon scattering amplitude in the convention where the incoming particle carrying k_h is the faster one.

Theorem 1 (soft magnon theorem, two-body). *Fix a compact $I \subseteq (0, \pi)$. For $k_h \in I$ and $k_s \rightarrow 0$,*

$$\delta(k_s, k_h) = 2k_s + \frac{v(k_h)}{\omega(k_h)} k_s^2 + O(k_s^3) \quad (1)$$

uniformly on I ; reversing which particle is soft flips the sign of the linear term and nothing else. Equivalently, the connected amplitude factorizes, $M_2(k_s, k_h) = 2ik_s M_1(k_h) + R$ with $\|R\| \leq C_I k_s^2 \|M_1\|$.

The zeroth order, $S_{12} \rightarrow 1$, is Dyson’s decoupling [9]. The content of the theorem is the next two orders. The linear coefficient is the pure number 2: independent of the hard momentum, of the coupling, of

everything. Hard-leg data enters first at second order, through the single invariant v_h/ω_h . The 2 is not a scattering length—the ferromagnet’s scattering length vanishes [15]—it is a *displacement*: $d\delta/dk_s = 2$ states that the soft packet exits the collision shifted by two lattice sites. We derive it without solving the model, from two ingredients the symmetry supplies: the operator identity $[H, Q_k] = (e^{ik} - 1)J_k$ obeyed by the deformed broken charge $Q_k = \sum_x e^{ikx} S_x^-$ and its current J_k , and the single contact term of the exchange interaction. The residue that fixes the coefficient is a current matrix element, $\langle k_h | Q_0^\dagger J_0 | k_h \rangle = 2i v(k_h)$: charge times velocity, the lattice image of the continuum soft-current analysis [3, 12]. The closed Bethe form of S_{12} enters only as verification; the hypotheses are symmetry, injectivity, and finite range, not integrability.

Three independent checks agree (Fig. 2). The closed-form expansion reproduces Eq. (1) exactly. Exact diagonalization on rings of 18–24 sites reproduces the two-magnon eigenvectors to 10^{-15} and the two coefficients to 10^{-10} . And real-time wavepacket collisions [30] measure the soft packet’s displacement directly: it matches $-d\delta/dk_s$ computed from the exact amplitude to parts in 10^5 at clean kinematics, for both signs of k_s .

We conjecture the general statement—Conjecture S: for scattering states with n hard legs on any injective MPS vacuum with broken continuous symmetry, $M_{n+1} = 2ik_s M_n + o(k_s)$ with the same coefficient. What separates theorem from conjecture is stated precisely in the Supplemental Material as two lemmas: uniform regularity of current form factors under wavepacket smearing, and an LSZ-type reduction showing the charge descendant dominates the soft amplitude. We have proved neither in general; both hold explicitly in the two-body sector.

Memory.—Memory needs a wall, so change the model minimally: the easy-axis XXZ ferromagnet, anisotropy $\Delta > 1$, with its two product vacua $|\cdots \uparrow\uparrow \cdots\rangle, |\cdots \downarrow\downarrow \cdots\rangle$ and a domain wall between them. Send in a magnon wavepacket with central momentum k_0 and ask where the wall sits after the collision (Fig. 3). Read the wall position from a window W around it, $\mathfrak{X}_W = \text{const} + \frac{1}{2s} \sum_{x \in W} (S_x^z + s)$. This observable changes only through the $U(1)$ currents on the two boundary bonds of W —an exact operator identity—so its total change over the event is the zero-frequency weight of the boundary current. Memory is the DC component of a current at the edge of the observation window: the lattice image of the continuum statement that memory is stored in the potential at the boundary [2, 7]. Charge bookkeeping then gives the memory law.

Theorem 2 (memory law). *Assume the state before and after the collision decomposes into a wall in W plus free magnon content outside it (hypothesis H-AD, Supplemen-*

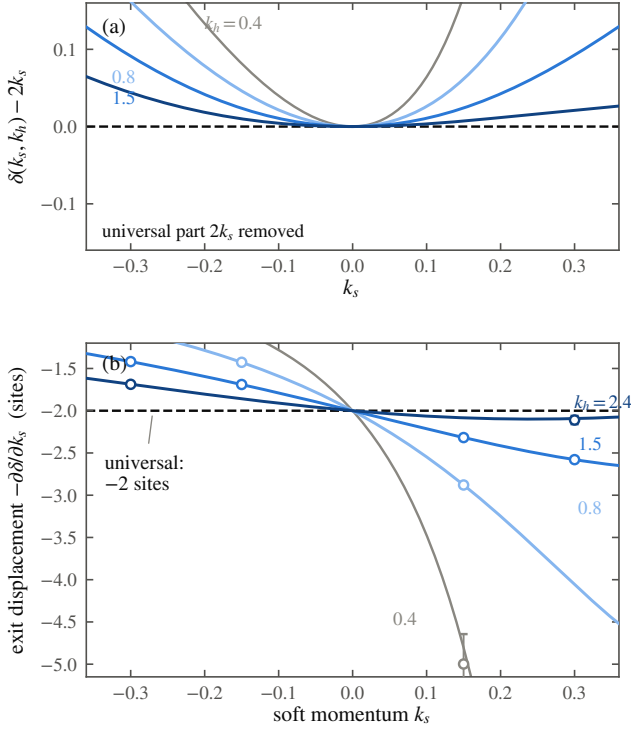


FIG. 2. The soft magnon theorem and its verifications. (a) The exact two-magnon phase of the Heisenberg ferromagnet with its universal part removed: $\delta(k_s, k_h) - 2k_s$ vanishes quadratically for every hard momentum k_h — the hard leg enters only at order k_s^2 , with coefficient v_h/ω_h , Eq. (1) (coefficients confirmed by exact diagonalization to 10^{-10}). (b) The displacement reading of the same statement: the soft packet’s exit displacement $-\partial\delta/\partial k_s$ for all k_h (curves, exact) passes through -2 sites at $k_s = 0$. Circles: real-time wavepacket measurements, agreeing with the exact curves to parts in 10^5 at clean kinematics (error bars mostly smaller than the symbols). The gray curve ($k_h = 0.4$) shows the non-uniformity of the soft expansion as $k_h \rightarrow 0$; the universal -2 survives it. Every soft magnon leaves a two-site footprint, whatever it scatters off.

tal Material). Then

$$\delta x = -\frac{1}{s} \langle N_T \rangle = -\frac{1}{s} \int \frac{dk}{2\pi} |\varphi(k)|^2 T(k), \quad (2)$$

where T is the wall’s transmission probability and φ the packet: each transmitted magnon displaces the wall by exactly $-1/s$ sites—two sites for spin $\frac{1}{2}$ —and each reflected one by zero.

The displacement is quantized. It does not depend on k_0 , on Δ , on the packet shape, or on any scattering phase; those enter only through $T(k)$. Sparse-sector numerics on chains of up to 200 sites confirm $\delta x = -2 \langle N_T \rangle$ to 0.004 sites, with six-figure convergence in system size. The transmission itself we obtain in closed form in the dominant domain-wall sector, where the kink–magnon

problem reduces to a side-coupled (Fano) level: $t(k) = [1 + iJ^2/4\omega(k)v(k)]^{-1}$ with $\omega(k) = J(\Delta - \cos k)$, matching the measured reflection over $\Delta \in [1.5, 12]$ to a few percent with no fitted parameter. As $k_0 \rightarrow 0$,

$$T(k_0) = 16(\Delta - 1)^2 k_0^2 + O(k_0^4) : \quad (3)$$

a soft magnon is totally reflected and leaves no memory. The memory effect has an Adler zero of its own. One warning earned by this analysis: the naive transcription of the continuum memory formula—displacement equals the DC limit of the soft factor—is *false* here; the wall’s quantum $1/s$ is a charge, not a phase, and the soft data enters only through T . And one observation we can state but not yet derive: the two 2’s of this Letter—the soft displacement $d\delta/dk_s = 2$ and the memory quantum $1/s = 2$ —are numerically the same and dimensionally the same object, the $U(1)$ charge transported past an observer in units of the vacuum spin density. We conjecture they are one invariant; the spin-1 chain, where the memory quantum becomes one site, will falsify or confirm this.

Finally, the third edge closes at the level of bookkeeping: δx records which ordered pair of vacua the chain interpolates between before and after the event, which is exactly the data labelling the orbit of the broken asymptotic symmetry. Measuring the memory identifies the asymptotic transformation that acted.

The SPT index in soft amplitudes.—Does the central extension $[\omega]$ of the asymptotic charge algebra show up in scattering? Our preliminary analysis points to a dichotomy, which we state as a conjecture: in the bulk, $[\omega]$ cancels from soft amplitudes at every order, because only the adjoint action of $V(g)$ enters there; at an open edge or a symmetry twist a single unpaired $V(g)$ survives, and the soft response then detects $[\omega]$ —through degeneracies and selection rules rather than through modified expansion coefficients. This is consistent with the continuum principle that anomalies constrain soft limits [31, 32] and with the anomalous-edge analysis of MPO symmetries [28, 29, 33]; the bulk half is within reach of the methods above, and we expect to settle it.

Outlook.—Three directions are concrete. The spin-1 chain tests the two-2’s conjecture, since the memory quantum halves while the soft phase can be computed independently. Higher dimensions replace bond matrices by the virtual symmetries of projected entangled-pair states, where the same interior cancellation applies. And magnonic and cold-atom platforms already resolve single-site wall displacements and host gravitational-wave analogues [16, 34]; a two-site kick per transmitted magnon is an observable, not a metaphor. The asymmetry of our opening—magnon and wall as unrelated objects—has disappeared: the wall is the broken asymptotic symmetry made visible, and the magnon’s soft phase and the wall’s memory are the same conserved charge read out in two ways. The infrared structure of gauge theory and

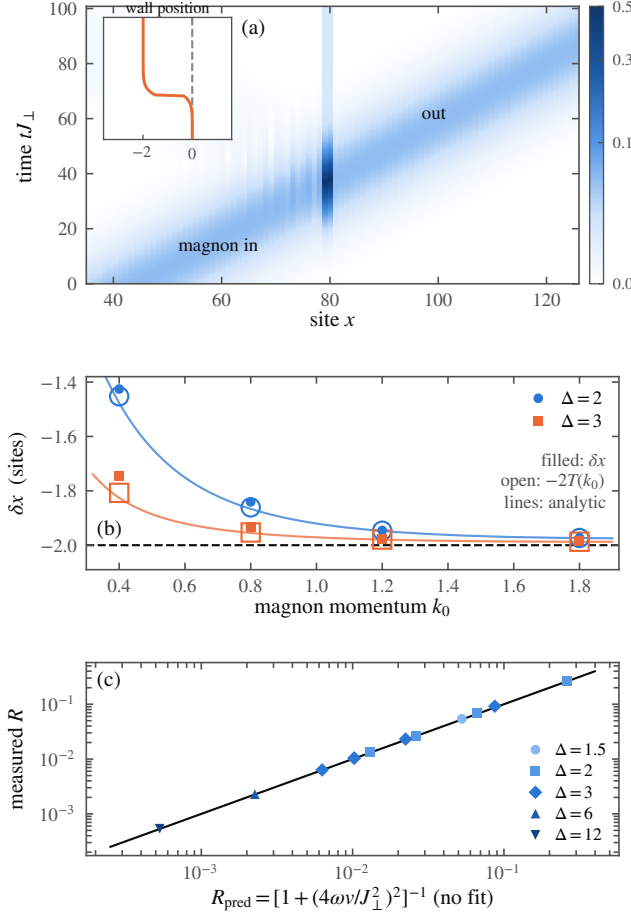


FIG. 3. The quantized memory effect in the XXZ chain ($\Delta > 1$). (a) Space-time record of one event ($\Delta = 3$, $k_0 = 1.2$, $N = 160$): flipped-spin density $\frac{1}{2} - |\langle S_x^z \rangle|$ as a magnon wavepacket crosses the wall. Inset: the wall position (half-magnetization crossing), which steps back by two sites during the collision and stays there — the permanent record. Across the scan the displacement obeys $\delta x = -2T(k_0)$ to 0.004 sites on chains of up to 200 sites. (b) The measured reflection $R = 1 - T$ across $\Delta \in [1.5, 12]$ and $k_0 \in [0.4, 1.8]$ collapses onto the parameter-free Fano formula $R = [1 + (4\omega v/J^2)^2]^{-1}$ (curves) to a few percent. As $k_0 \rightarrow 0$ the transmission vanishes quadratically, $T = 16(\Delta - 1)^2 k_0^2$: a soft magnon leaves no memory—the memory effect's own Adler zero. The displacement per transmission is exactly two sites: a quantized, permanent, zero-frequency record of the scattering event, readable arbitrarily long after the magnon has gone.

gravity, long tied to null infinity, fits on a chain of spins, where every step of it can be checked.

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END MATTER

The Supplemental Material contains: (i) the proof of the truncated-symmetry identity and of the asymptotic

charge algebra $\mathbb{C}_\omega[G]$, with the strong-convergence dichotomy; (ii) the kink construction as a half-infinite symmetry string and the vacuum-pair classification; (iii) the proof of Theorem 1, its current-algebra derivation, and the two lemmas separating it from Conjecture S; (iv) the Bethe-ansatz oracle computation with remainder bounds, and the exceptional-state inventory behind the completeness statement; (v) the proof of Theorem 2 and the hypothesis H-AD; (vi) the Fano reduction for $t(k)$ and its validity domain; (vii) numerical methods and the machine-checked certificates for every theorem.

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